

Online Consent Form

The Use of Conversational Agents in Microeconomic Learning

Principal Investigator Name and Title: Dr. Tiffany (Wenting) Li

Department and Institution: Department of Computer Science, Stevens Institute of Technology

Contact Information: wli69@stevens.edu, 201.216.3751

Summary

You are being asked to participate in a voluntary research study. The purpose of the study is to understand how learners leverage chatbots as they learn microeconomics. The study should take approximately 30 minutes. Risks are minimal; participation in this study does not entail any cognitive, physical, or emotional risks beyond those of everyday life. By participating in this study, you will not only learn something new in microeconomics, but also help us design better conversational agents that help future learners engage in more effective learning.

What procedures are involved?

Participating in this study will first involve answering 8 multiple-choice questions on microeconomics. Next, you will be asked to work on the same 8 problems again and understand the concepts behind them. You will have access to a regular state-of-the-art generative chatbot (similar to ChatGPT) in the process. You will receive feedback upon submitting each problem. Finally, you will answer a few demographic questions. Lastly, you will receive a completion code in the survey and be asked to paste it into the Prolific task page.

What potential risks are involved?

The risks of the study are minimal. This study may use advice or information created by a computer program called artificial intelligence (AI). Sometimes AI can make mistakes, leave out important details, or exhibit unfair bias. The research team will check the information, but there is still a small chance that something it produces may be wrong or not fit your situation. These risks are similar to those you encounter when using state-of-the-art commercial chatbots (such as ChatGPT and Gemini). AI advice should not replace help from a trained professional. If anything is confusing or doesn't seem right, you can contact the researchers at any time.

You may also feel unsure when faced with unfamiliar concepts, nervous about taking a test, and upset when you get problems wrong. These risks are similar to those you experience when learning new materials or skills at school or at work. If you feel the emotions above, you can contact the researcher, who will point you to available resources to help.

What potential benefits are there?

There are no direct benefits to you from taking part in this study. But as you participate, you may learn something new about Microeconomics. And by participating in the study, you can help us develop a better understanding of how learners use conversational agents during learning. With this knowledge, we can improve the current design of chatbots to better support future learners.

Will I be reimbursed for any expenses or paid for my participation in this research?

Online Consent Form

You will receive an \$8 payment for completing the study.

Can I withdraw or be removed from the study?

If you decide to participate, you are free to withdraw your consent by contacting the researchers and discontinue participation at any time. Your participation in this research is voluntary. We will delete your data once you withdraw your consent. Your decision whether or not to participate, or to withdraw after beginning participation, will not affect your performance statistics on Prolific.

How will the researchers protect my information?

In this study, the only identifier we collect from you is your platform identifier. It will be stored separately from the rest of the research data, deleted permanently after 3 years, and will not be published or presented.

Who will have access to the information collected during this research study?

Efforts will be made to limit the use and disclosure of your personal information, including research study records, to those who have a need to review it. We cannot promise complete secrecy. There are reasons why information about you may be used or seen by other people beyond the research team during or after this study. For example, university officials, government officials, study funders, auditors, and the Institutional Review Board may need access to study information to ensure the study is conducted safely and appropriately.

How might the information collected in this study be shared in the future?

We will retain the de-identified information we collect about you for recordkeeping purposes. But the information will not be shared with anyone outside of the research team.

Who should I contact if I have questions?

If you have any questions about this study or your part in it, or if you have concerns or complaints about the research, please feel free to contact any of the researchers:

- Eiliyah Sarowar at esarowar@stevens.edu
- Tiffany Li at wli69@stevens.edu

What are my rights as a research subject?

If you have any questions about your rights as a research subject, including concerns, complaints, or to offer input, you may email the Stevens Institutional Review Board (IRB) at IRB@stevens.edu. You will have the option to provide feedback or concerns anonymously or you may provide your name and contact information for follow-up purposes.

Please print this consent form if you would like to retain a copy for your records.

- ☐ I am 18 years or older.
- ☐ I am currently physically located in the United States.
- ☐ I have read and understood the above information.
- ☐ I voluntarily agree to participate in this research.